ELEC 221 Lecture 13
CT Fourier transform: convolution, multiplication; the magnitude-phase representation; step response

Thursday 22 October 2026

Announcements

Monday 2 Nov. tutorial

- TA3 due tomorrow 23:59
- Monday is last tutorial before midterm (midterm prep)
- Midterm exam cutoff is next Thursday's class
- Quiz 7 Tuesday
- Final exam time announced: Friday 18 December 15:30-18:00
- Tomorrow's office hours on Zoom

Last time

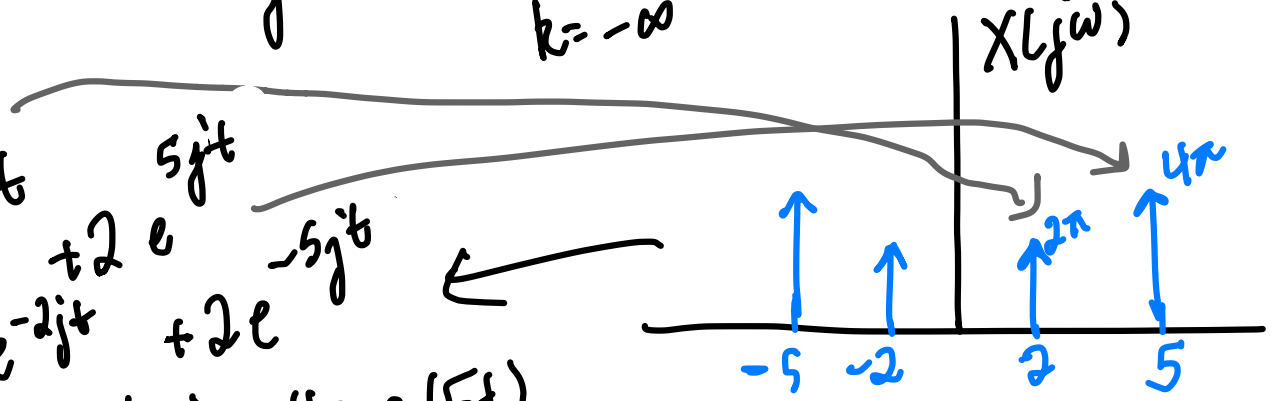
We evaluated Fourier spectra for *periodic* functions.

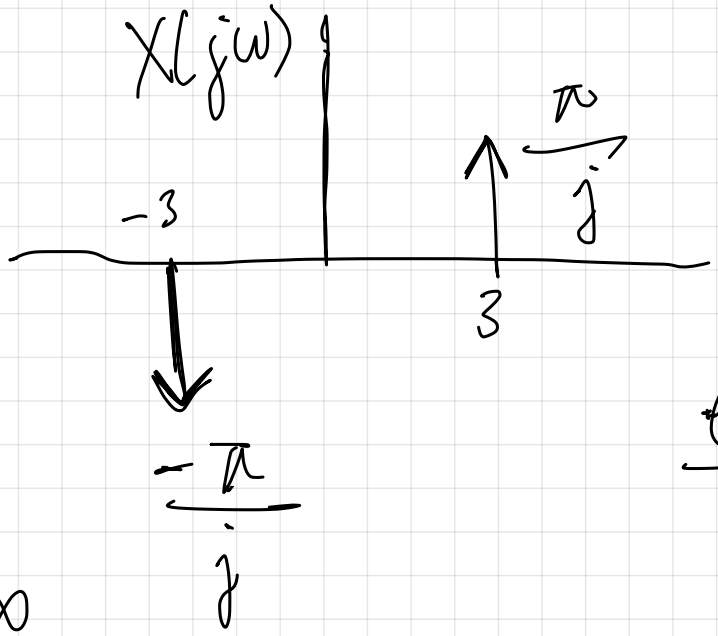
$$x(t) = \sum_{k=-\infty}^{\infty} c_k e^{jk\omega t}$$

$$X(j\omega) = \sum_{k=-\infty}^{\infty} 2\pi c_k \delta(\omega - k\omega_0)$$

$$x(t) = e^{2jt} + 2e^{5jt} + e^{-2jt} + 2e^{-5jt}$$

$$= 2\cos(2t) + 4\cos(5t)$$





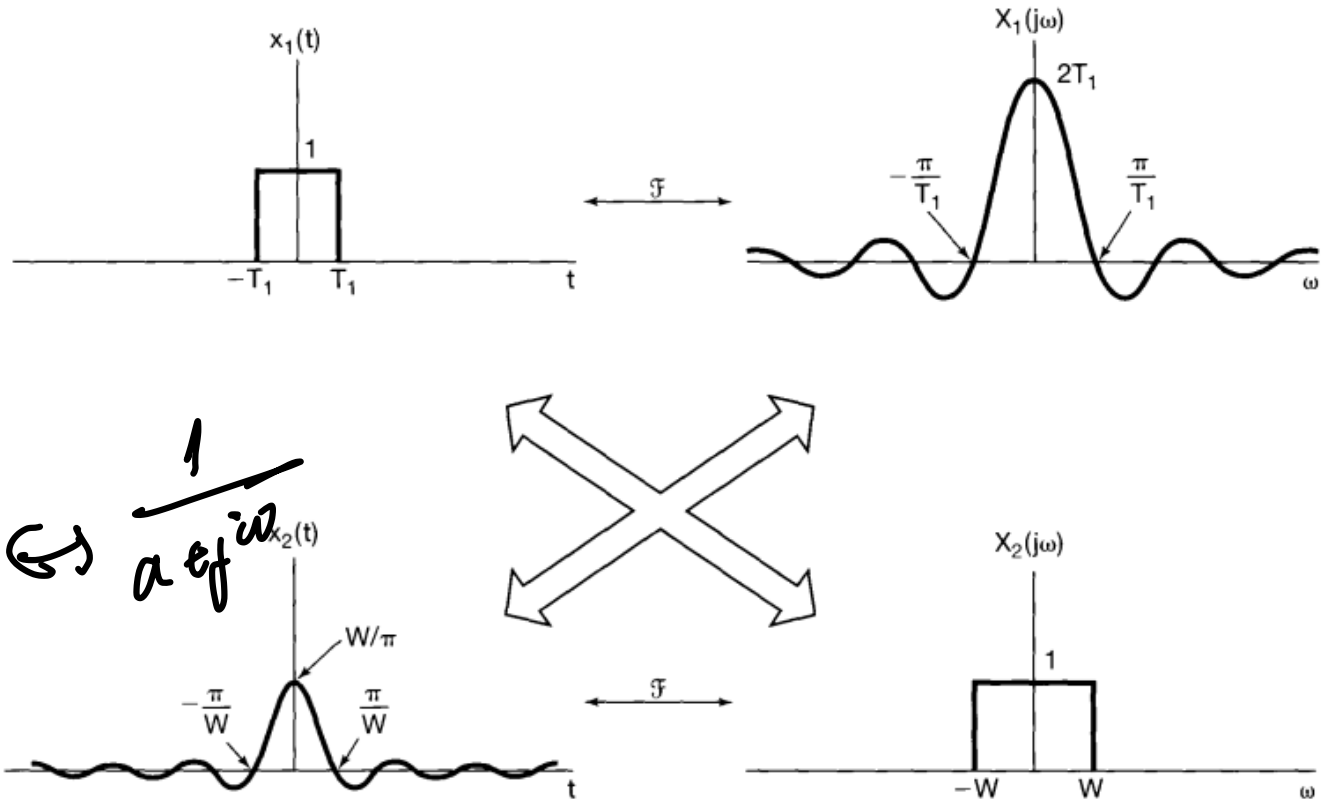
$$\frac{e^{3jt} - e^{-3jt}}{2j}$$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{\pi}{j} \left[\delta(\omega - 3) - \delta(\omega + 3) \right] e^{j\omega t} d\omega$$

$$= \frac{1}{2j} \left[e^{j3t} - e^{-j3t} \right] = \sin(3t)$$

Last time

We observed time-frequency duality in the Fourier transform.



$e^{-at}u(t) \leftrightarrow \frac{1}{a+j\omega}$

Image credit: Oppenheim chapter 4.3

Last time

$$y(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} Y(j\omega)\, e^{j\omega t}\, d\omega = \underbrace{\frac{1}{2\pi}\int_{-\infty}^{\infty} H(j\omega)\, X(j\omega)\, e^{j\omega t}\, d\omega}_{\text{from conv. integral}}$$

We derived a very important property of Fourier transforms under *convolution*:

$$y(t) = h(t) * x(t) \quad \Longleftrightarrow \quad Y(j\omega) = H(j\omega)\, X(j\omega)$$

This makes it easier to analyze LTI systems!

Today

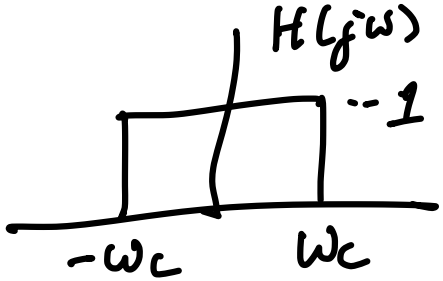
Learning outcomes:

- Apply the *convolution property* of the Fourier transform to characterize LTI system behaviour
- Describe the *multiplication property* of the Fourier transform and provide applications of its use
- Express a Fourier transform using the magnitude-phase representation and read a Bode plot
- Compute the step response of LTI systems described by first-order ODEs

Example: convolution

Suppose $x(t) = \frac{\sin(\omega_0 t)}{\pi t}$ is input into a lowpass filter

$$H(j\omega) = \begin{cases} 1 & |\omega| < \omega_c \\ 0 & |\omega| \geq \omega_c \end{cases}$$



"original way":

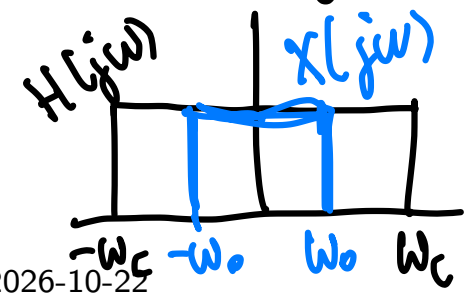
$$y(t) = h(t) * x(t) \qquad h(t) = \frac{\sin(\omega_c t)}{\pi t}$$

$$= \int_{-\infty}^{\infty} h(\tau)\, x(t-\tau)\, d\tau$$

$$= \int_{-\infty}^{\infty} \frac{\sin(\omega_c \tau)}{\pi \tau} \cdot \frac{\sin(\omega_0 (t-\tau))}{\pi (t-\tau)}\, d\tau = \text{!!}$$

Better way:

$$Y(j\omega) = H(j\omega) \cdot X(j\omega)$$



$$Y(j\omega) = \begin{cases} 1 & |\omega| < \omega_0 \\ 0 & \text{otherwise} \end{cases}$$

$\omega_c > \omega_0$

$$x(t) = \frac{\sin(\omega_0 t)}{\pi t} \Rightarrow X(j\omega) = \begin{cases} 1 & |\omega| < \omega_0 \\ 0 & |\omega| > \omega_0 \end{cases}$$

☆ work out $\omega_c < \omega_0$ case yourself.

$$\Rightarrow y(t) = \frac{\sin(\omega_0 t)}{\pi t}$$

Exercise: convolution





Consider an LTI system that sends

$$x(t) = e^{-t}u(t) \rightarrow y(t) = \frac{1}{2}e^{-|t|}$$

$$= \frac{1}{2}e^{-t}u(t) + \frac{1}{2}e^{t}u(-t)$$

What is its impulse response?

$$Y(j\omega) = H(j\omega)X(j\omega)$$

$$\frac{Y(j\omega)}{X(j\omega)} = H(j\omega)$$

→ Inverse Fourier transform for $h(t)$

☆ Table 4.1!

$$x(t) = e^{-t}u(t)$$

$$X(j\omega) = \frac{1}{1+j\omega}$$

$$y(t) = \frac{1}{2}e^{-|t|} = \frac{1}{2}e^{-t}u(t) + \frac{1}{2}e^{t}u(-t)$$

$$Y(j\omega) = \frac{1}{2}\frac{1}{1+j\omega} + \frac{1}{2}\frac{1}{1-j\omega}$$

Exercise: convolution

$$H(j\omega) = \frac{Y(j\omega)}{X(j\omega)} = \frac{\frac{1}{2}\frac{1}{1+j\omega} + \frac{1}{2}\frac{1}{1-j\omega}}{\frac{1}{1+j\omega}}$$

$$= \frac{1}{2} + \frac{1}{2}\frac{1}{1-j\omega} \cdot 1+j\omega$$

$$= \frac{1}{2}\left[1 + \frac{1+j\omega}{1-j\omega}\right]$$

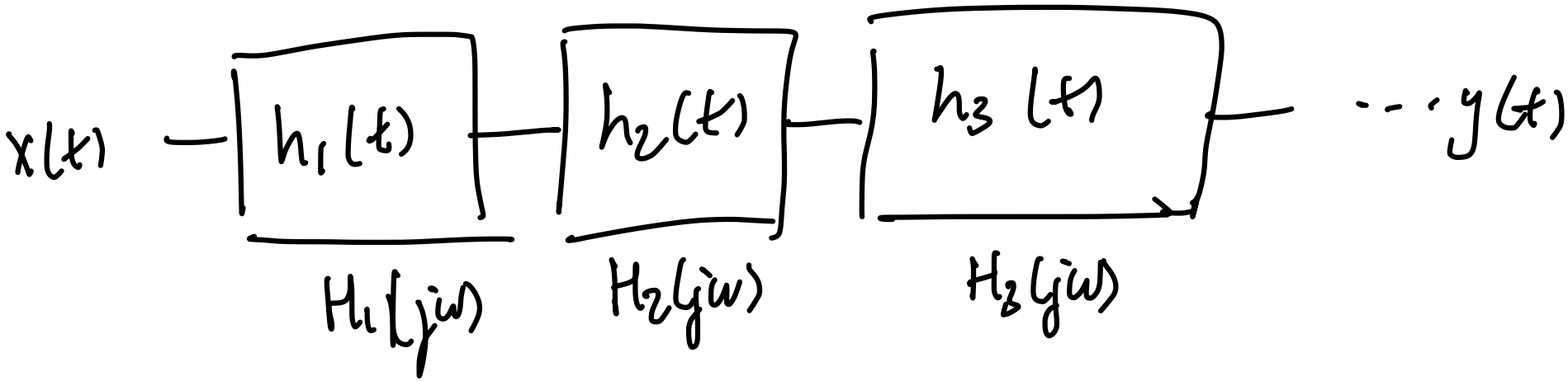
$$= \dots$$

$$= \frac{1}{1-j\omega}$$

$$\Rightarrow h(t) = e^{t}\,u(-t)$$

Intuition: cascade of systems

What is the frequency response of the following setup?



$$\Rightarrow H(j\omega) = H_1(j\omega) H_2(j\omega) H_3(j\omega) \quad \checkmark$$

$$h(t) = h_1(t) * h_2(t) * h_3(t) \quad \int$$

also works but more tedious.

The multiplication property

We know that:

$$y(t) = h(t) * x(t)$$

$$Y(j\omega) = H(j\omega)\, X(j\omega)$$

Something similar holds when we interchange time and frequency:

$$r(t) = s(t)\, p(t)$$

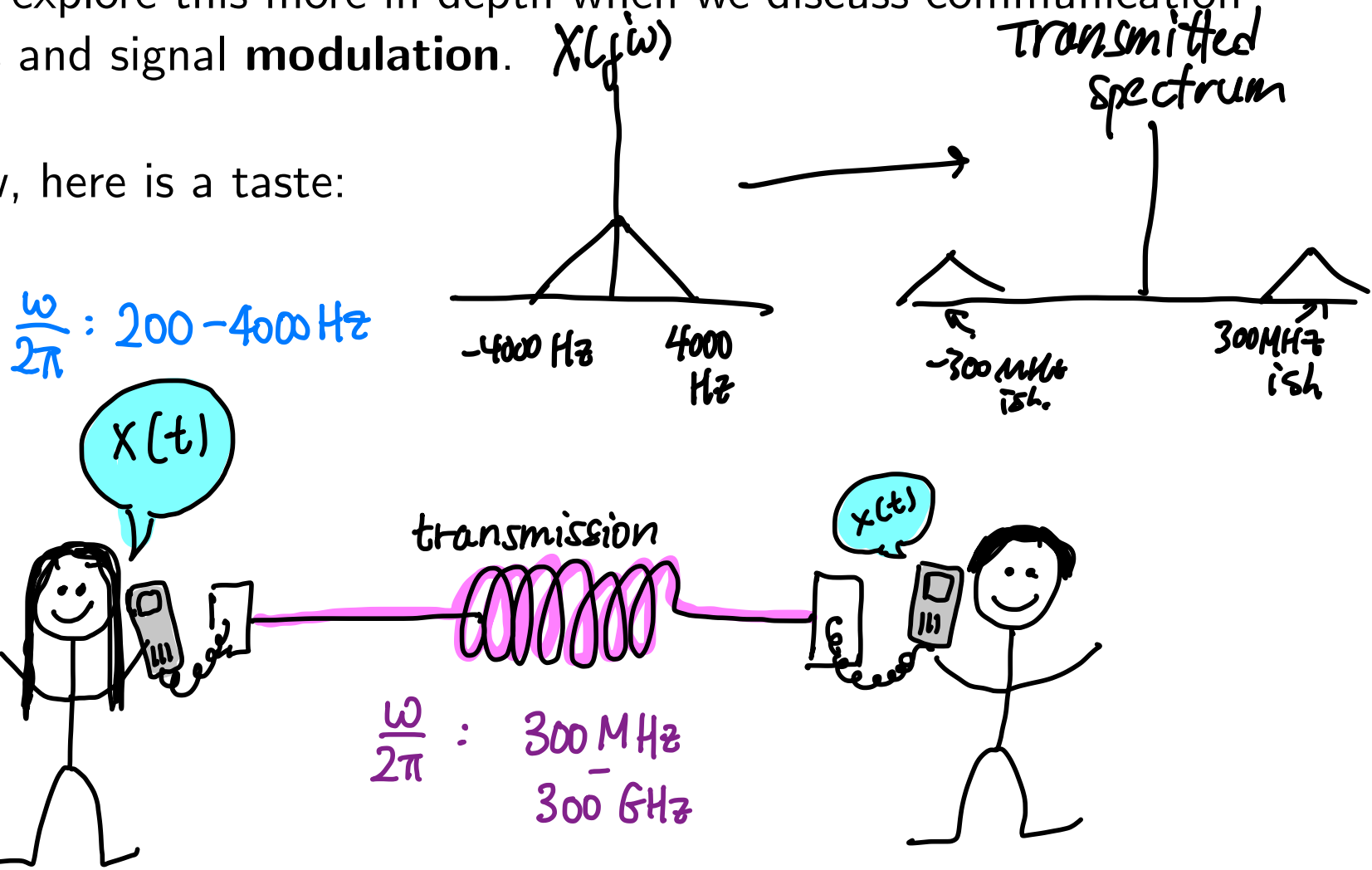
$$R(j\omega) = S(j\omega) * P(j\omega)$$

This is the **multiplication property**.

Example: the multiplication property

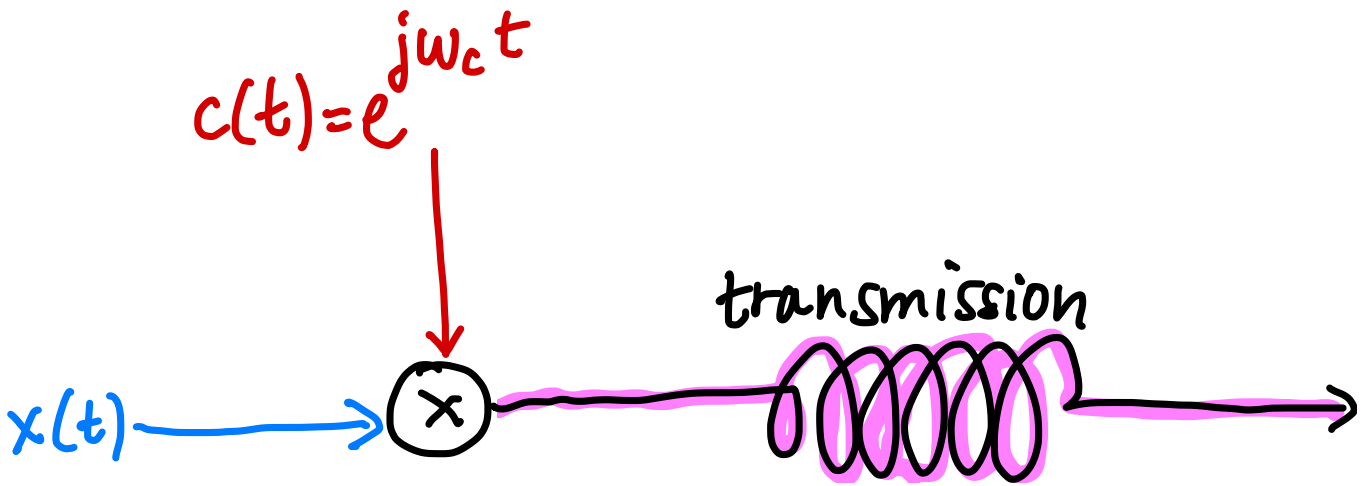
We will explore this more in depth when we discuss communication systems and signal **modulation**.

For now, here is a taste:



Example: the multiplication property

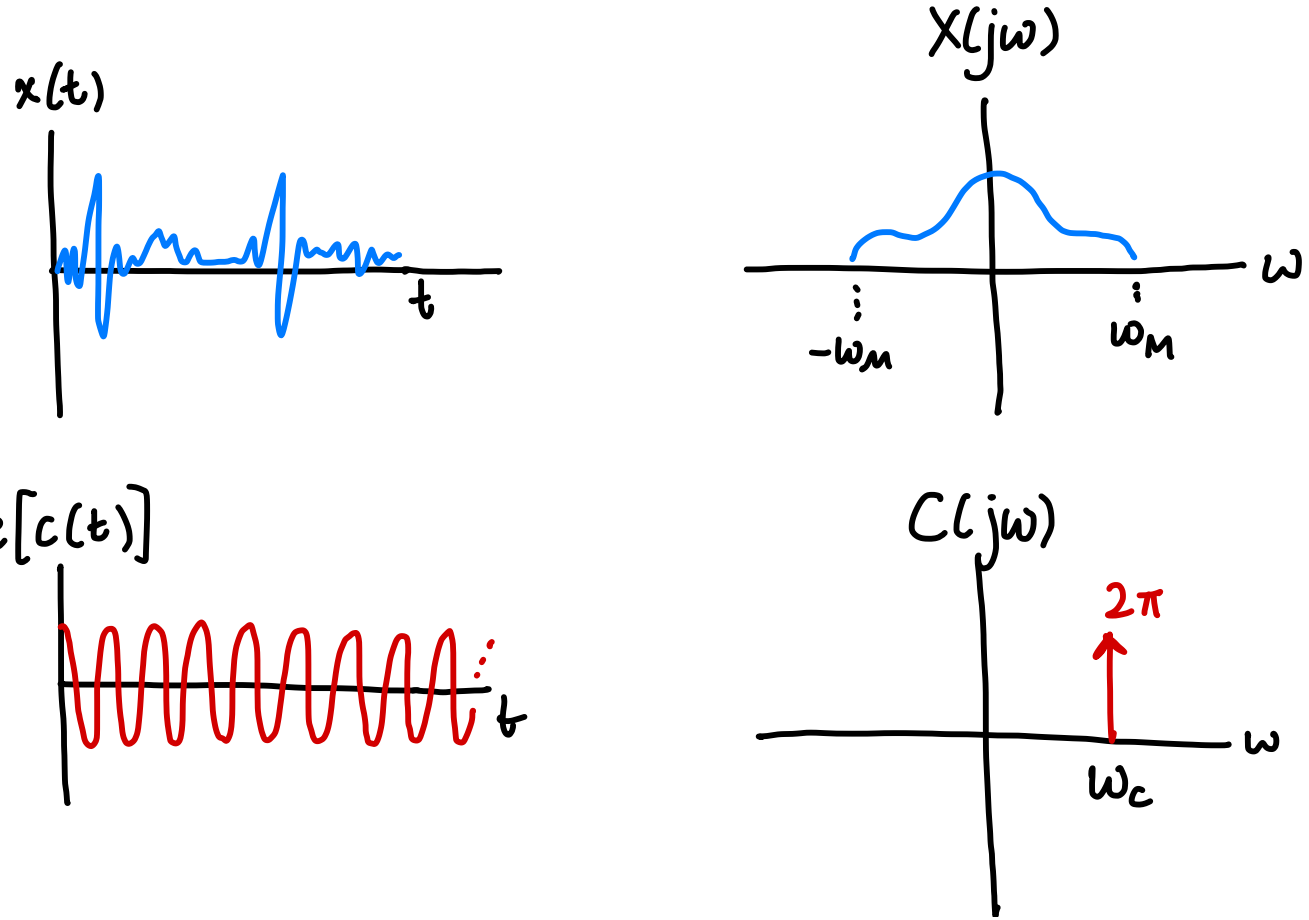
To shift our signal into the frequency range of transmission, we can multiply it by a **carrier signal** (amplitude modulation):



Is this doing what we think it is?

Example: the multiplication property

Consider the Fourier spectrum of both signals:



Example: the multiplication property

The multiplication property tells us

$$y(t) = x(t)\,c(t)$$

$$Y(j\omega) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(j\theta)\,C(j(\omega-\theta))\,d\theta$$

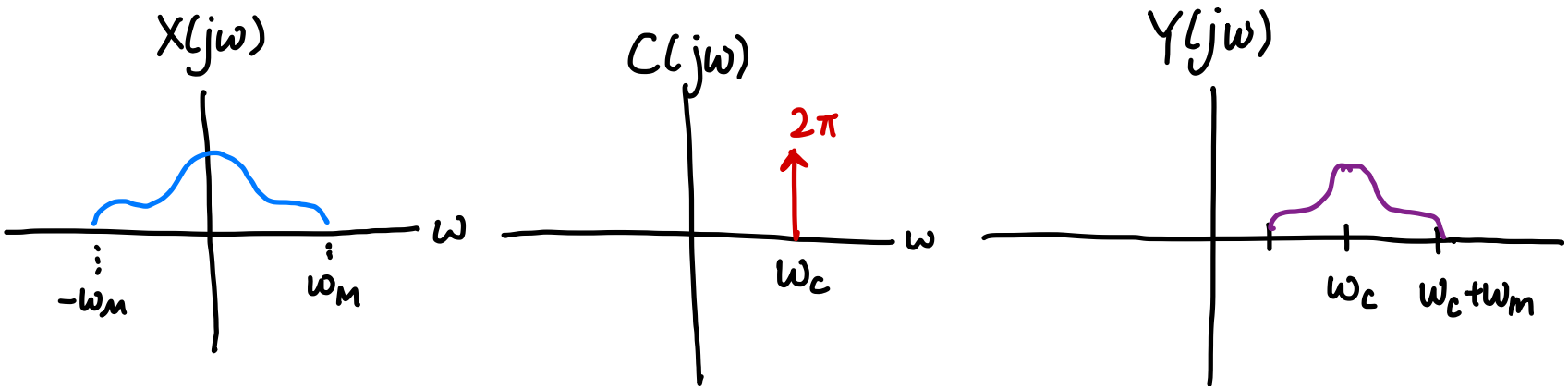
We have $C(j\omega) = 2\pi\,\delta(\omega-\omega_c)$

$$Y(j\omega) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(j\theta)\cdot 2\pi\,\delta((\omega-\omega_c)-\theta)\,d\theta$$

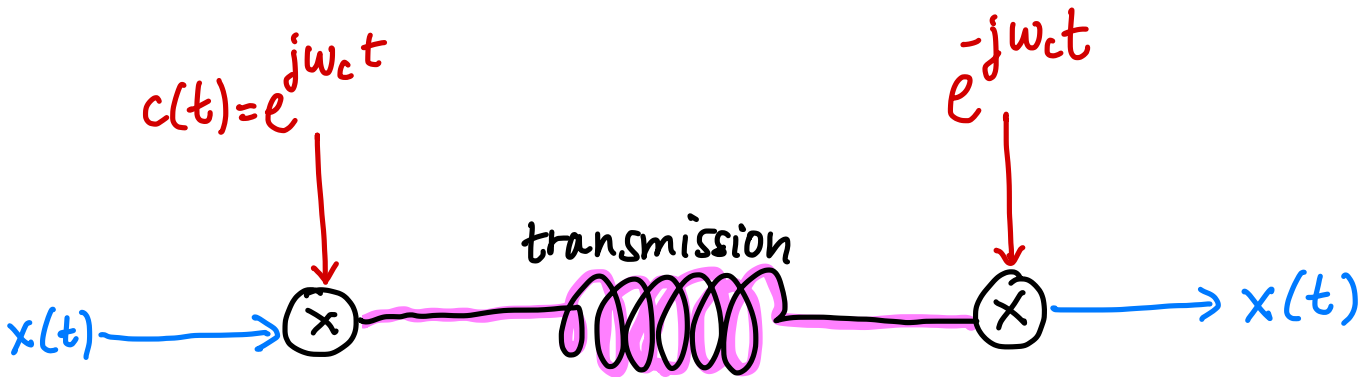
$$= X(j(\omega-\omega_c))$$

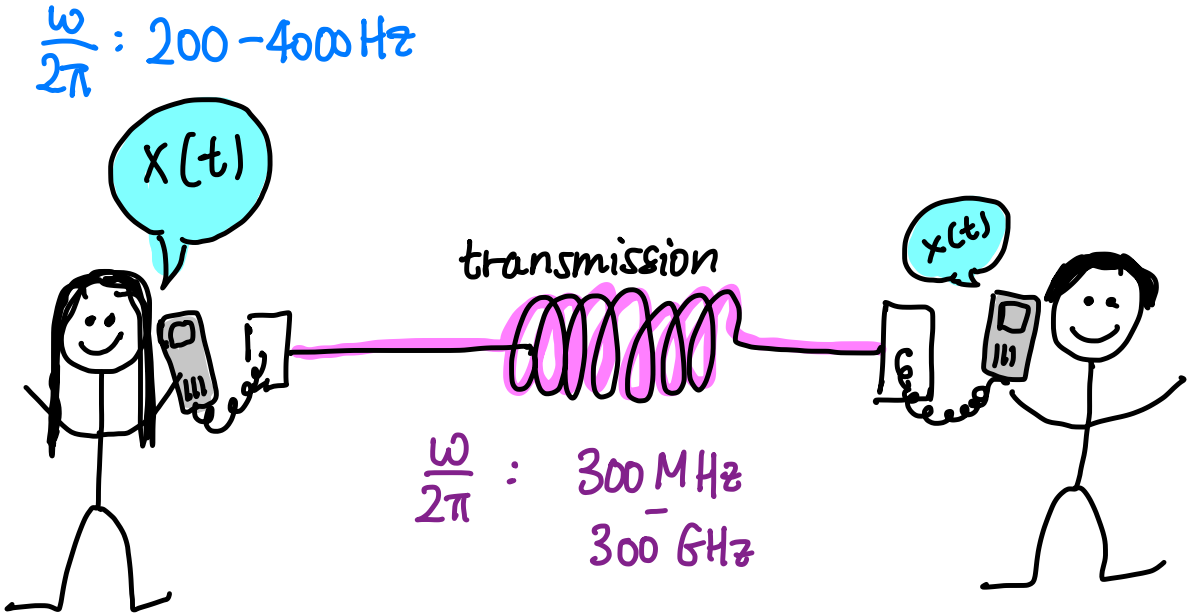
Example: the multiplication property

Multiplication with complex exponential carrier signal shifts the spectrum. We can move it into the desired frequency range.



Example: the multiplication property





Example: frequency-selective filtering with variable centre frequency

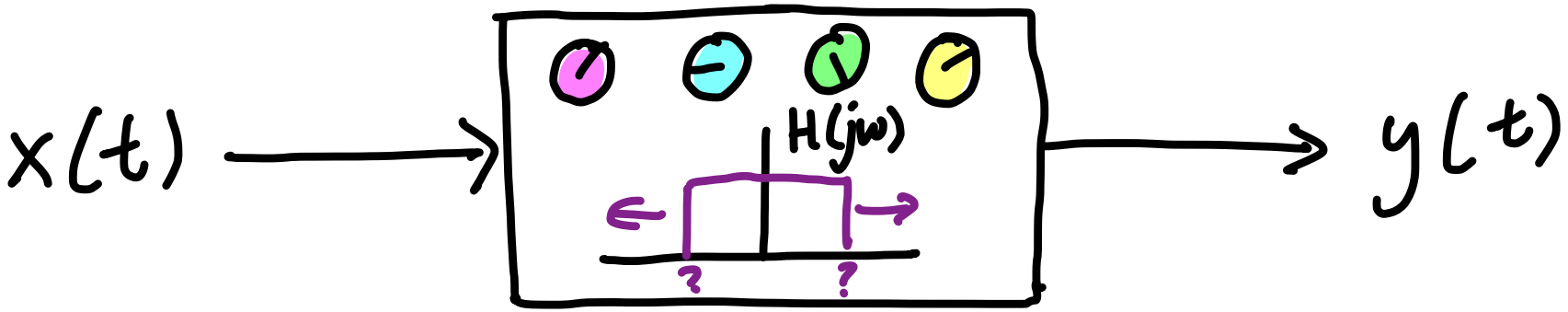
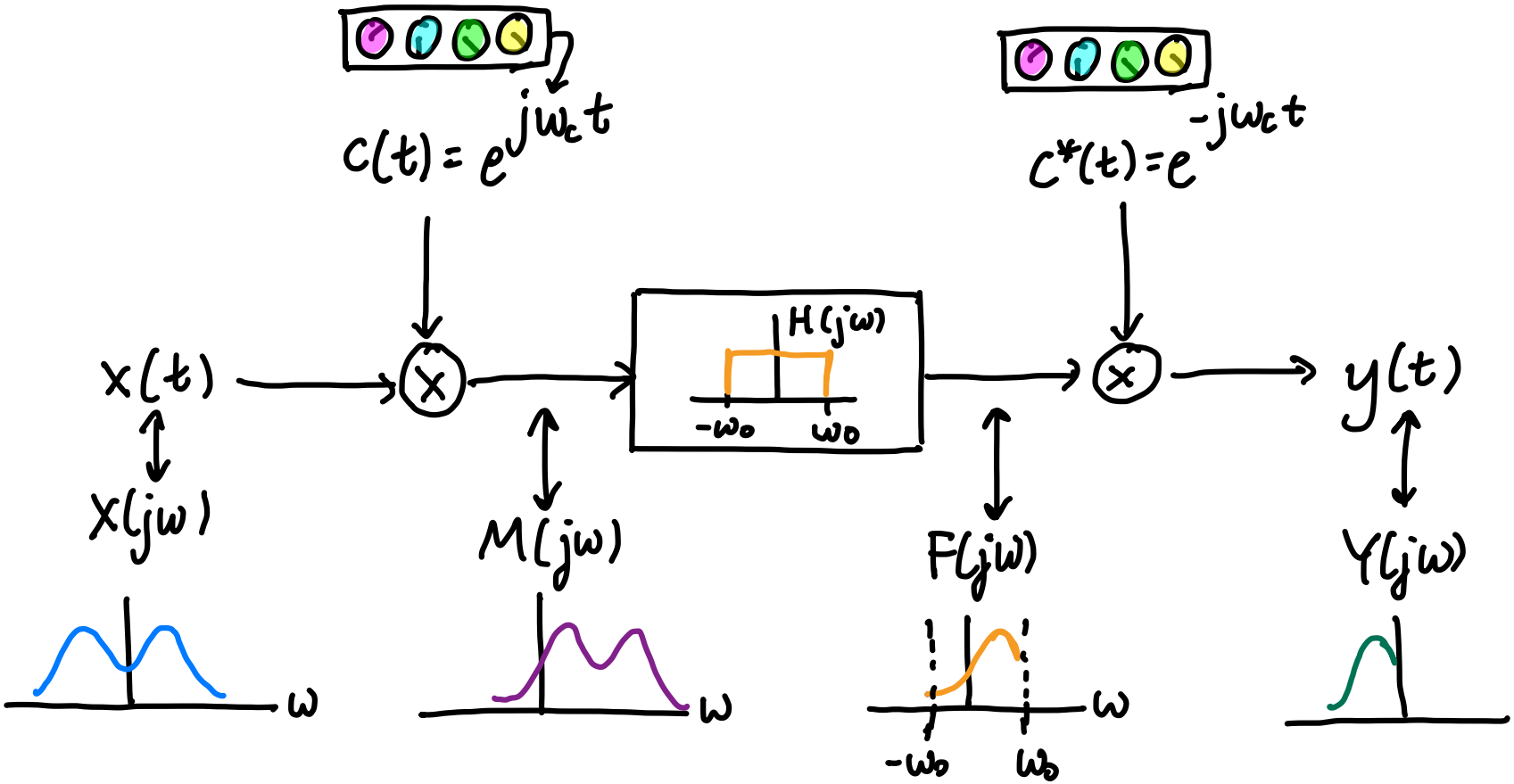




Image: `https://www.euronics.ee/UserFiles/Products/Images/162391-panasonic-rf-2400d-radio.png`

Example: frequency-selective filtering with variable centre frequency



The magnitude-phase representation

what is $|X(j\omega)|$ for $\delta(\omega)$?

Since Fourier spectra are complex we can express them in terms of their magnitude and phase.

$$X(j\omega) = |X(j\omega)| \, e^{j \angle X(j\omega)}$$

magnitude; phase

Recall the convolution property:

$$Y(j\omega) = H(j\omega) \cdot X(j\omega)$$

Exercise: How does a system $H(j\omega)$ affect $|X(j\omega)|$ and $\angle X(j\omega)$?

$$|Y(j\omega)| = |H(j\omega)| \cdot |X(j\omega)|$$

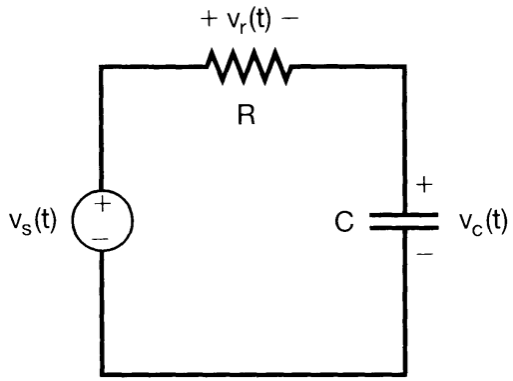
gain

$$\angle Y(j\omega) = \angle H(j\omega) + \angle X(j\omega)$$

↳ phase shift.

Example: Lowpass filters in practice

The magnitude-phase representation helps visualize and characterize systems. Recall our RC circuit / lowpass filter.



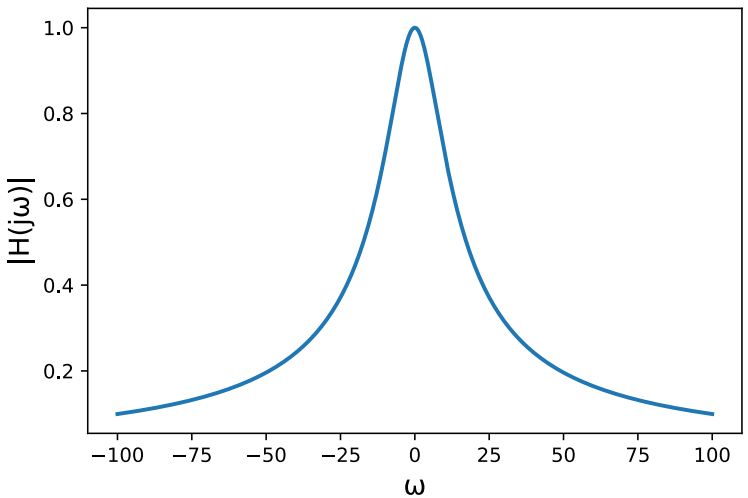
We wrote a differential equation for the system and found

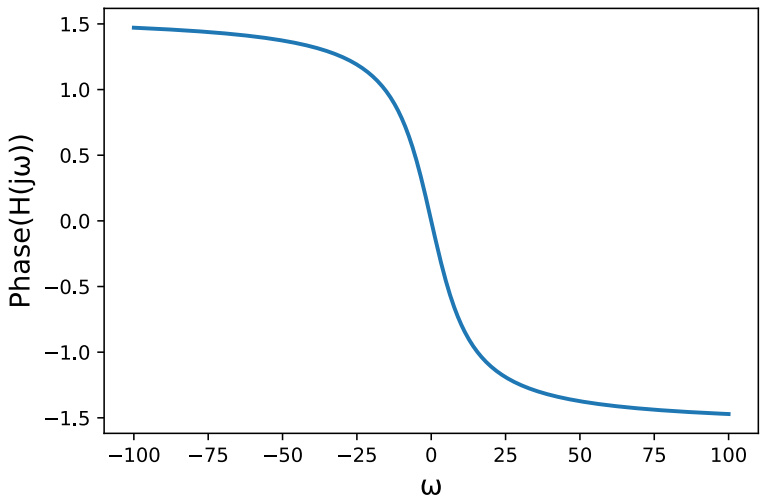
$$RC\frac{dv_c(t)}{dt} + v_c(t) = v_s(t)$$

$$H(j\omega) = \frac{1}{1 + j\omega RC}$$

Example: lowpass filters in practice

Results in the following frequency response (setting $RC = 0.1$):





Adjusting RC controls the frequency response; the phase shift is also frequency dependent.

Bode plots

Recall:

$$|Y(j\omega)| = |H(j\omega)||X(j\omega)|$$

$$\angle Y(j\omega) = \angle H(j\omega) + \angle X(j\omega)$$

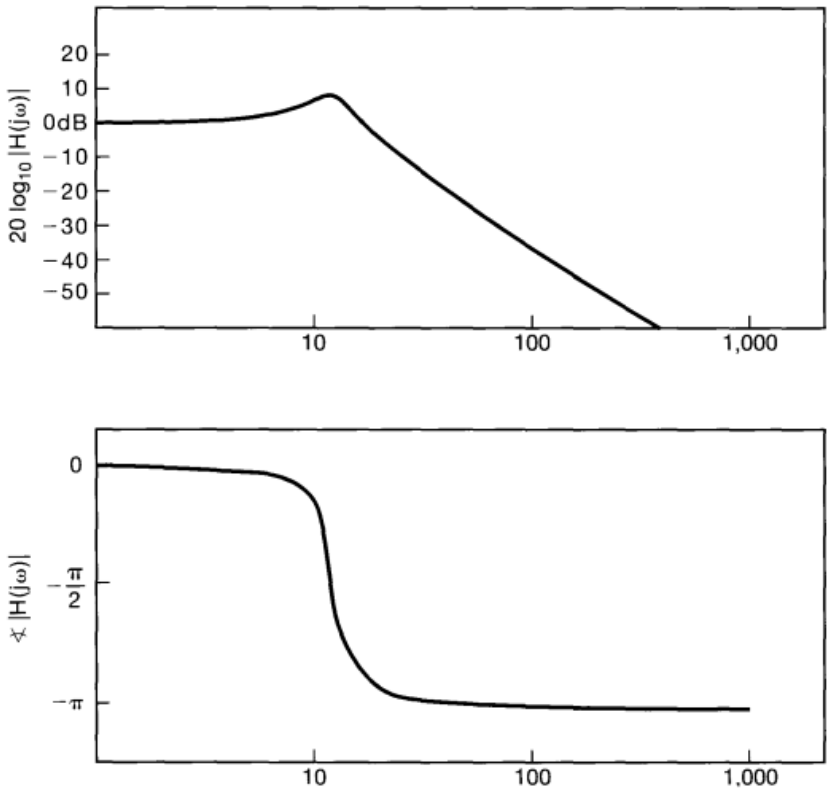
Magnitude is multiplicative and phase is additive... would be nicer if both were additive.

$$\log|Y(j\omega)| = \log|H(j\omega)| + \log|X(j\omega)|$$

Rather than plotting $|H(j\omega)|$ and $\angle H(j\omega)$, it is common to plot $20\log_{10}|H(j\omega)|$ and $\angle H(j\omega)$ against $\log_{10}\omega$.

Bode plots

These are called *Bode plots*:



The logarithmic scale also allows us to view the response over a much wider range of frequencies.

Image credit: Oppenheim 6.2

For next time

Content:

- CT Fourier transform under differentiation and integration
- Systems described by second-order ODEs

Action items:

1. Tutorial assignment 3
2. Prepare for Quiz 7
3. Study for midterm

Recommended reading:

- For today's class: Oppenheim 4.3-4.7, 6.1-6.2
- Suggested problems: 4.5, 4.9, 4.11-4.16, 4.19, 4.22, 4.25, 4.26, 4.29, 6.1, 6.3a, 6.5, 6.9
- For next class: Oppenheim 4.7, 6.3-6.5